MILITARY STANDARD

MIL-STD-1320-229A (NAVY)
16 MAR 1989

SUPERSEDING
MIL-STD-1320-229
15 NOVEMBER 1988

TRUCKLOADING

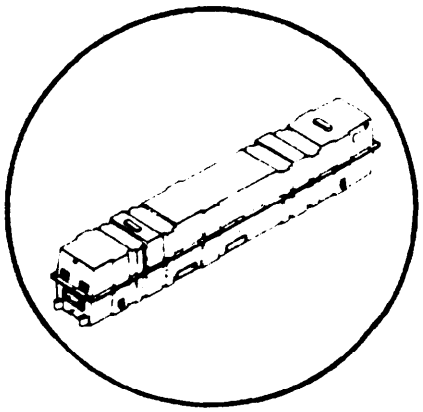
STANDARD MISSILE BOOSTER

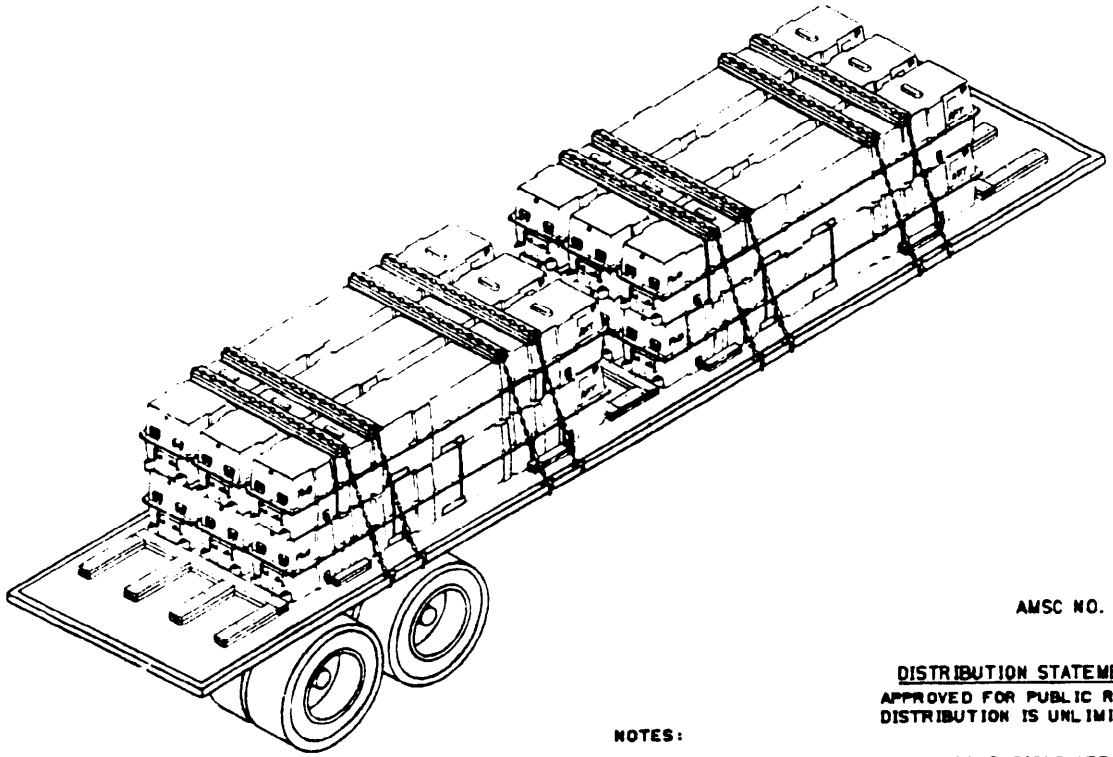
IN SHIPPING AND STORAGE CONTAINER, MK 686

CONTAINER DATA

DODIC/NALC
BOOSTERS PER CONTAINER 1
DIMENSIONS . . . . . 178 L X 27 W X 30 H (SH)
GROSS WEIGHT . . . . . 2759 LBS Δ
TARE WEIGHT . . . . . 529 LBS Δ
CUBE . . . . . 83.4 CU FT
Δ DO NOT USE FOR SHIPPING WEIGHT

DOT HAZARD CLASSIFICATION
BOOSTER . . . . . EXPLOSIVES B
EMPTY CONTAINER . . . INERT





AMSC NO. - N/A

DISTRIBUTION STATEMENT A
APPROVED FOR PUBLIC RELEASE: DISTRIBUTION IS UNLIMITED.

NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
			APPROVAL	
A	ADD UNITIZING PROCEDURE	3/16/89	ER	XT

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY.

Robert A. Ronald 11-13-88
SIGNATURE TDA, WPNSTA EARLE DATE

E. Kelly 11-15-88
SIGNATURE SEASYSCOM BY DIRECTION DATE

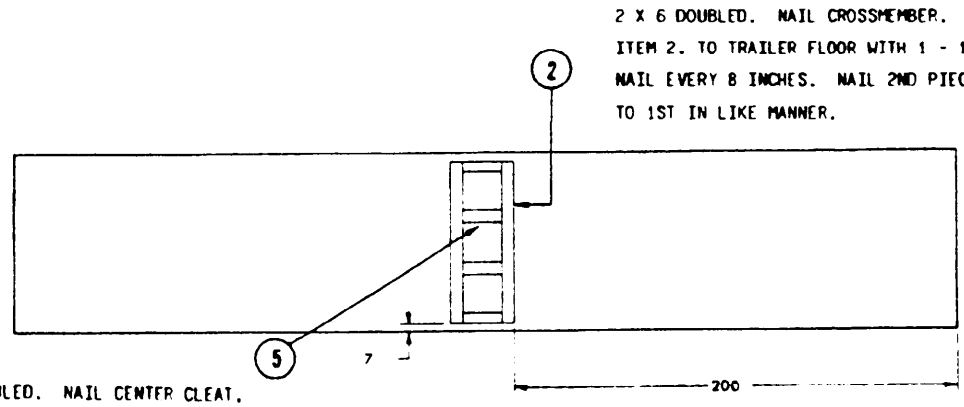
ORIGINATOR A. W. Kretchmer 4/12/88
DATE

WPNSTA EARLE, NEW JERSEY

Document Provided by: https://assist.dla.mil -- Downloaded: 20260410-23123-48Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING STANDARD MISSILE BOOSTER IN SHIPPING AND STORAGE CONTAINER MK 68G. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY).
2. THE PROCEDURES DESCRIBED HEREIN ARE INTENDED FOR 40 FT AND LONGER FLATBED TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS. ADJUST DIMENSIONS FOR LONGER AND/OR WIDER TRAILERS. DO NOT USE SHORTER TRAILERS TO SHIP TWO STACKS.
3. CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE-TO-ONE BASIS FOR TIE DOWNS.
4. STEEL STRAPPING USED AS TIE DOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
5. CHAINS, FITTINGS AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY) EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL).
6. PROTECTOR BOARDS (DOUBLED 2 X 6'S) SHALL BE USED UNDER THE TIE DOWNS. WHEN USING STEEL STRAPPING, THE STRAPS SHALL BE STAPLED TO THE PROTECTOR BOARD USING 2 INCH STRAPPING STAPLES. WHEN USING CHAIN, THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARD BY DRIVING A 10 NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIEDOWN.
7. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION, REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
8. UNLESS OTHERWISE SPECIFIED DUNNAGE LUMBER, STEEL STRAPPING, STRAP SEALS AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
9. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
10. AFTER THE BLOCKING, BRACING AND TIE DOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT, WATERPROOF TARPAULINS.
11. AFTER THE TARPAULINS ARE IN PLACE, ATTACH APPROPRIATE PLACARDS TO BOTH SIDES, FRONT AND BACK OF THE TRAILER (WHEN APPLICABLE) AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA ON THE BACK DECK OF THE TRAILER.



2 X 6 DOUBLED. NAIL CROSSMEMBER, ITEM 2, TO TRAILER FLOOR WITH 1 - 16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

2 X 6 DOUBLED. NAIL CENTER CLEAT, ITEM 5, TO TRAILER FLOOR WITH 4 - 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

DETAIL A

PREPOSITIONED FLOOR BLOCKING FOR 40 FT OR LONGER TRAILER.

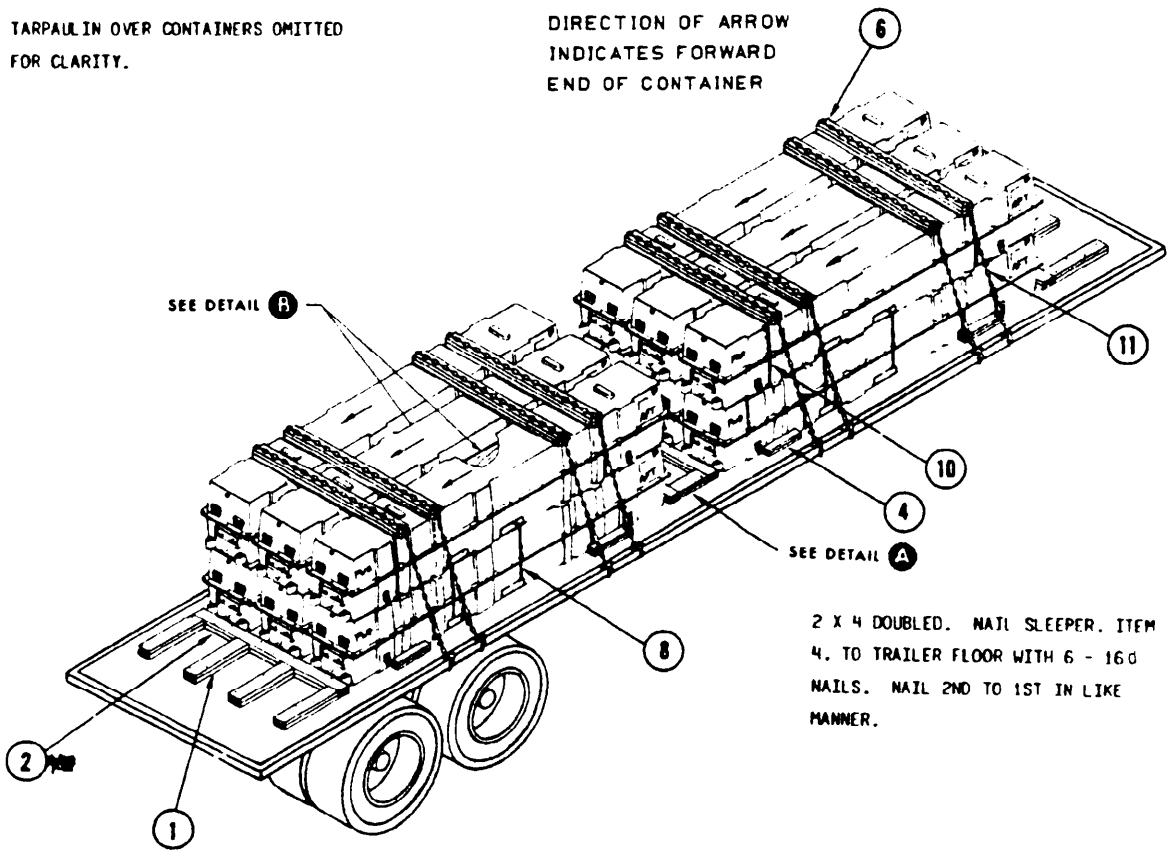
Source: https://assist.dla.mil -- Downloaded: 202610-23T23:48Z
Check the source to verify that this is the current version before use.

FTL 40 FT AND LONGER TRAILER (FLATBED)

NOTE

TARPAULIN OVER CONTAINERS OMITTED FOR CLARITY.

DIRECTION OF ARROW INDICATES FORWARD END OF CONTAINER



2 X 4 DOUBLED. NAIL SLEEPER. ITEM 4. TO TRAILER FLOOR WITH 6 - 16d NAILS. NAIL 2ND TO 1ST IN LIKE MANNER.

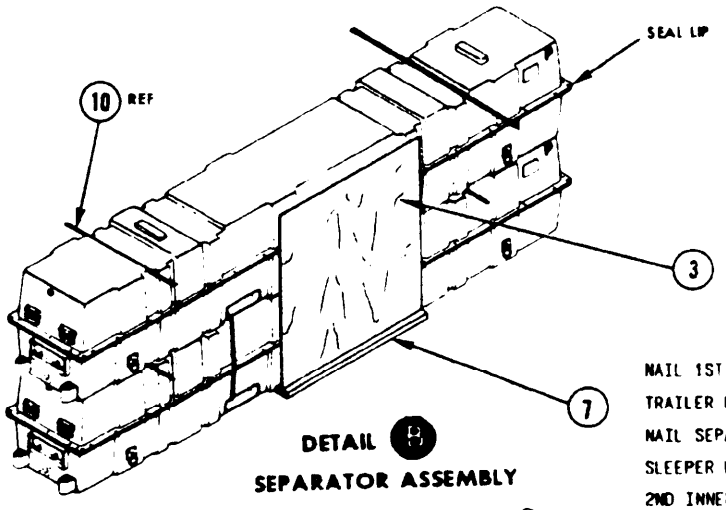
2 X 6 DOUBLED. NAIL BACKUP CLEAT. ITEM 1. TO TRAILER FLOOR WITH 9 - 16d NAILS. NAIL 2ND TO 1ST IN LIKE MANNER.

QUANTITY SHOWN IS FOR FTL. REDUCE QUANTITY AS REQUIRED FOR LTL.

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
8	11	CHAIN AND LOAD BINDER	SEE NOTE 5, PAGE 2	3/8 OR 5/16
4	10	CROSS STRAP	SEE NOTE 8, PAGE 2	1-1/4 X .035
16	9	SEAL, NOTCHING	SEE NOTE 8, PAGE 2	1-1/4
12	8	STRAP	SEE NOTE 8, PAGE 2	1-1/4 X .035
8	7	SLEEPER, INNER	SEE NOTE 8, PAGE 2 (WOOD)	2 X 2 X 48
16	6	PROTECTOR BOARD	SEE NOTE 8, PAGE 2 (WOOD)	2 X 6 X 81
8	5	CENTER CLEAT	SEE NOTE 8, PAGE 2 (WOOD)	2 X 6 X 18
16	4	SLEEPER	SEE NOTE 8, PAGE 2 (WOOD)	2 X 4 X 24
4	3	SEPARATOR, PLYWOOD	SEE NOTE 8, PAGE 2	3/8 X 48 X 57
8	2	CROSSMEMBER	SEE NOTE 8, PAGE 2 (WOOD)	2 X 6 X 82
16	1	BACKUP CLEAT	SEE NOTE 8, PAGE 2 (WOOD)	2 X 6 X 30

LIST OF MATERIALS

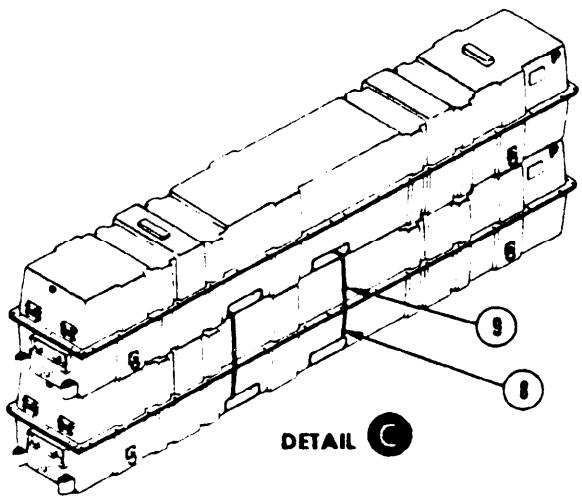
Source: https://assist.dla.mil -- Downloaded: 2025-10-05T20:44:47
Check the source to verify that this is the current version before use.



DETAIL B
SEPARATOR ASSEMBLY

FOR DOUBLE LAYER
USE 3/8 X 48 X 57
FOR SINGLE LAYER
USE 3/8 X 48 X 26

NAIL 1ST INNER SLEEPER, ITEM 7, TO TRAILER FLOOR WITH 4 - 16d NAILS. NAIL SEPARATOR, ITEM 3, TO INNER SLEEPER WITH 4 - 16d NAILS. NAIL 2ND INNER SLEEPER TO TRAILER FLOOR WITH 4 - 16d NAILS.



DETAIL C

UNITIZING PROCEDURE

NOTE

BE SURE TO ALIGN THE STACKING FEATURES.

1. USING A FORKTRUCK, STACK TWO CONTAINERS TOGETHER.
2. STRAP THE TOP CONTAINER TO THE BOTTOM CONTAINER THROUGH EACH SET A RESPECTIVE FORK POCKET USING STEEL STRAPPING, ITEM 8. TENSION AND DOUBLE NOTCH SEAL, ITEM 9.

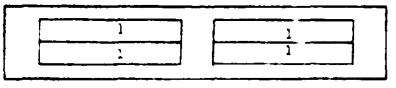
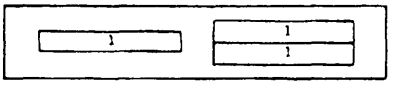
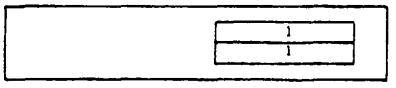
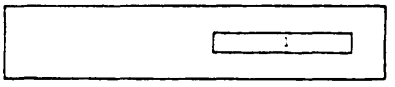
WARNING

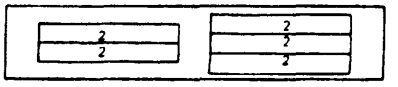
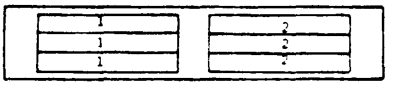
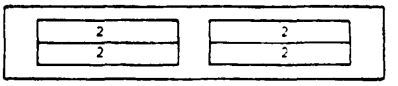
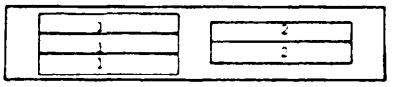
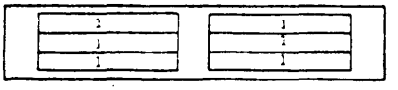
THE UNITIZING OF CONTAINERS TWO HIGH IS AUTHORIZED FOR LOADING AND UNLOADING OF TRAILERS ONLY. ANY MOVEMENTS OF THESE UNITIZED CONTAINERS IS LIMITED TO THAT NECESSARY TO LOAD OR UNLOAD THE TRAILER.

Source: https://assist.dla.mil -- Downloaded: 20260-23T23:48Z
Check the source to verify that this is the current version before use.

LTL 40 FT AND LONGER TRAILER (FLATBED)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF CONTAINERS TO BE SHIPPED FROM TABLE 1.
2. POSITION CONTAINERS ON THE TRAILER AS SHOWN IN TABLE 1. BLOCK AND TIE DOWN USING THE PRINCIPLES SHOWN IN THE GENERAL DOCUMENT MIL-STD-1320 (NAVY).
3. FOR SINGLE LAYER STACKS, USE 2 CHAIN AND LOAD BINDERS PER STACK.

NO. OF CONTAINERS	LOAD PATTERN
5	
4	
3	
2	
1	

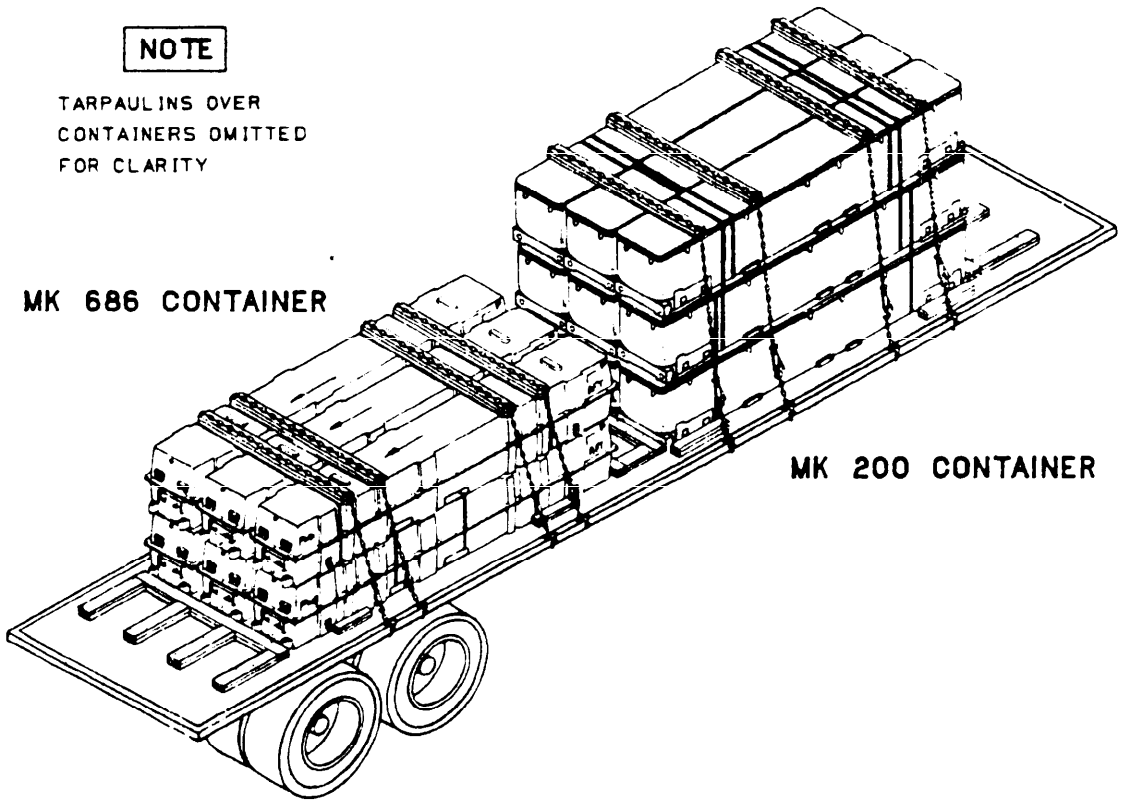
NO. OF CONTAINERS	LOAD PATTERN
10	
9	
8	
7	
6	

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:48Z
Check the source to verify that this is the current version before use.

MIXED TRUCKLOAD

NOTE

TARPAULINS OVER CONTAINERS OMITTED FOR CLARITY



NOTES:

1. A MIXED TRUCKLOAD OF BOOSTERS IN MK 686 OR MK 200 CONTAINERS CAN BE ACCOMPLISHED BY REPLACING THE FRONT STACK OF MK 686 CONTAINERS WITH A STACK OF MK 200 CONTAINERS AS SHOWN.
2. UNITIZING PROCEDURES FOR MK 200 CONTAINERS ARE CONTAINED IN MIL-STD-1320-18.

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N816)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-STD-1320-229A STANDARD Missile Booster in Container MK 686

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.

OLD

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
ENALTY FOR PRIVATE USE $300

Commanding Officer
Naval Weapons Station Earle
Code 804 - M. Shelly
Colts Neck, New Jersey 07722-5000

OLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.